

eg 1

1)

$$y = \sqrt{\frac{x^2}{x-1}} \quad \text{funzione composta, } f(g(x)) = f'(g(x)) \cdot g'(x)$$

$$\left(\frac{x^2}{x-1}\right)^{1/2} = \frac{1}{2} \left(\frac{x^2}{x-1}\right)^{1/2-1} = \frac{1}{2} \left(\frac{x^2}{x-1}\right)^{-1/2} = \frac{1}{2 \sqrt{\frac{x^2}{x-1}}}$$

$$\frac{x^2}{x-1} = \frac{(2x)(x-1) - (x^2)(1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2}{(x-1)^2} = \frac{x^2 - 2x}{x-1}$$

$$f'(x) = \frac{1}{2 \sqrt{\frac{x^2}{x-1}}} \cdot \frac{x^2 - 2x}{x-1} = \frac{x^2 - 2x}{2 \sqrt{\frac{x^2}{x-1}}}$$

2)

$$f(x) = \frac{\sqrt{x} \lg(x)}{\lg(x)+1} \quad \frac{d}{dx} \sqrt{x} \lg(x) = \left(\frac{1}{2\sqrt{x}}\right)(\lg x) + (\sqrt{x})\left(\frac{1}{x}\right) = \frac{\lg x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}$$

$$f'(x) = \frac{\left(\frac{\lg x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right)(\lg x + 1) - (\sqrt{x} \lg x)\left(\frac{1}{x}\right)}{(\lg x + 1)^2}$$

$$f'(x) = \frac{\left(\frac{\lg x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right)(\lg x + 1) - (\sqrt{x} \lg x)\left(\frac{1}{x}\right)}{(\lg x + 1)^2} = \frac{\left(\frac{\lg x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right)(\lg x + 1) - \frac{\sqrt{x} \lg x}{x}}{(\lg x + 1)^2}$$

$$\frac{\left(\frac{\lg x}{2\sqrt{x}} + \frac{\sqrt{x}}{x}\right)(\lg x + 1) - \frac{\sqrt{x} \lg x}{x}}{(\lg x + 1)^2} = \frac{\frac{\lg^2(x)}{2\sqrt{x}} + \frac{\lg(x)}{2\sqrt{x}} + \frac{\sqrt{x} \lg x}{x} + \frac{\sqrt{x}}{x} - \frac{\sqrt{x} \lg x}{x}}{(\lg(x) + 1)^2}$$

$$\frac{\frac{\lg^2(x)}{2\sqrt{x}} + \frac{\lg(x)}{2\sqrt{x}} + \frac{\sqrt{x}}{x}}{(\lg(x) + 1)^2}$$

3)

$$f(x) = \arctan(\arccos(x))$$

$$f'(x) = \frac{1}{1 + \arccos^2 x} \cdot \frac{-1}{\sqrt{1-x^2}}$$

$$f'(x) = -\frac{1}{(1 + \arccos^2(x))(\sqrt{1-x^2})}$$

$$f(x) = \arctan(\arccos(x))$$

$$f'(x) = \frac{1}{1 + \arccos^2 x}$$

$$f(x) = \arccos x$$

$$f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

4)

$$f(x) = \sin(\cos x) = e^{\cos x \cdot \ln(\sin x)}$$

$$f' = \sin x^{\cos x} \cdot \left(-\sin x \ln(\sin x) + \cos x \cdot \frac{\cos x}{\sin x} \right)$$

$$f(x) = e^{\cos x \cdot \ln(\sin x)}$$

$$f'(x) = e^{\cos x \cdot \ln(\sin x)}$$

$$f(x) = \cos x \cdot \ln(\sin x)$$

$$f'(x) = (-\sin x)(\ln(\sin x)) + (\cos x) \left(\frac{1}{\sin x} \cdot \cos x \right)$$

$$f'(x) = -\sin x \ln(\sin x) + \cos x \cdot \frac{\cos x}{\sin x}$$

5)

$$f(x) = \lg \left| \frac{x-1}{x-3} \right|$$

$$f(x) = \begin{cases} \lg \frac{x-1}{x-3} & \text{für } x < 1 \vee x > 3 \\ -\lg \frac{x-1}{x-3} & \text{für } 1 < x < 3 \end{cases}$$

1)

$$f(x) = \lg \frac{x-1}{x-3}$$

$$f(g(x)) = f'(g(x)) \cdot g'(x)$$

$$f'(x) = \frac{1}{\frac{x-1}{x-3}} \cdot \frac{(1)(x-3) - (x-1)(1)}{(x-3)^2} = \frac{1}{\frac{x-1}{x-3}} \cdot \frac{(x-3) - (x-1)}{(x-1)^2} = \frac{x-3}{x-1} \cdot \frac{\cancel{x-3} - \cancel{x} + 1}{(x-1)^2} = \frac{x-3}{x-1} \cdot \frac{-2}{(x-1)^2}$$

$$\frac{\cancel{x-3}}{x-1} \cdot \frac{-2}{(x-1)^2} = \frac{-2}{(x-1)(x-1)}$$

2)

$$f'(x) = \frac{2}{(x-3)(x-1)}$$

6) ? ? !

$$f(x) = z^{\frac{x^2+1}{x-1}} = e^{\frac{x^2+1}{x-1} \cdot \ln(z)}$$

$$\frac{d}{dx} \frac{x^2+1}{x-1} = \frac{(2x)(x-1) - (x^2+1)(1)}{(x-1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$$

$$f'(x) = e^{\frac{x^2+1}{x-1} \cdot \ln(z)} \cdot \frac{(x^2 - 2x - 1)}{(x-1)^2} (\ln(z))$$

7)

$$f(x) = \arctan\left(\frac{1}{x}\right) + \frac{2}{x}$$

$$\frac{d}{dx} \left(\frac{2}{x}\right) = \frac{(0)(x) - (2)(1)}{x^2} = -\frac{2}{x^2}$$

$$f'(x) = -\frac{1}{x^2+1} - \frac{2}{x^2}$$

$$\frac{d}{dx} \arctan\left(\frac{1}{x}\right) = \frac{1}{1 + \left(\frac{1}{x}\right)^2} \cdot \frac{0(x) - (1)(1)}{x^2} =$$

$$\frac{1}{1 + \frac{1}{x^2}} \cdot -\frac{1}{x^2} = -\frac{1}{\left(1 + \frac{1}{x^2}\right)(x^2)} = -\frac{1}{x^2+1}$$

8)

$$f(x) = 2 + \sin^2 |x|$$

$$f(x) = \begin{cases} 2 + \sin^2 x & \text{per } x > 0 \\ 2 + \sin^2 -x & \text{per } x < 0 \end{cases}$$

1)

$$f(x) = 2 + \sin^2 x$$

$$f'(x) = 2 \sin x \cdot \cos x = 2 \sin x \cos x = \sin 2x$$

2)

$$f = 2 + \sin(-x) = 2 - \sin^2 x$$

$$f' = -2 \sin x \cdot (-\cos x) = \sin(2x)$$

$$f'(x) = \begin{cases} \sin 2x & x > 0 \\ \sin 2x & x < 0 \end{cases}$$

9)

$$f(x) = x^3 + |x|^3$$

$$f(x) = \begin{cases} x^3 + x^3, & x > 0 \\ x^3 - x^3, & x < 0 \end{cases}$$

1)

$$+ f'(x) = 3x^2 + 3x^2 = 6x^2$$

2)

$$f'(x) = 3x^2 - 3x^2 = 0$$

$$f'(x) = \begin{cases} 6x^2, & x > 0 \\ 0, & x < 0 \end{cases}$$

10)

$$f(x) = \log(\cos^2 x)$$

$$f'(x) = \frac{1}{\cos^2 x} \cdot (2 \cos x \cdot -\sin x) = \frac{1}{\cos^2 x} \cdot -2 \cos x \sin x = -\frac{2 \sin x}{\cos x} = -2 \tan$$

11)

$$f(x) = 7\pi e^2$$

$$f'(x) = 0$$

12)

$$f(x) = \frac{1}{\sqrt[3]{\cos x}} = (\cos x)^{-1/3}$$

$$f'(x) = -\frac{1}{3} \cos^{-1/3-1} x \cdot -\sin x = -\frac{1}{3} \cos^{-4/3} x \cdot -\sin x = \frac{1}{3^3 \sqrt[3]{\cos^4 x}} \cdot -\sin x = \frac{-\sin x}{3^3 \sqrt[3]{\cos^4 x}}$$

13)

$$f(x) = x^x + \frac{\lg x}{x} + \frac{x}{\lg x}$$

$$f'(x) = (x^x + x^x \ln x) + \left(\frac{1 - \lg x}{x^2} \right) + \left(\frac{-1 + \lg x}{\lg^2 x} \right)$$

$$f(x) = x^x = e^{x \cdot \ln x}$$

$$f'(x) = e^{x \cdot \ln x} \cdot ((1)(\ln x) + (x \cdot \frac{1}{x}))$$

$$e^{x \cdot \ln x} \cdot (\ln x + 1) = x^x \cdot (\ln x + 1) = x^x + x^x \ln x$$

$$f(x) = \frac{\lg x}{x}$$

$$f'(x) = \frac{\left(\frac{1}{x}\right)(x) - (\lg x)}{x^2} = \frac{1 - \lg x}{x^2}$$

$$f(x) = \frac{x}{\lg x}$$

$$f'(x) = \frac{\lg x - (x)\left(\frac{1}{x}\right)}{\lg^2 x} = \frac{\lg x - 1}{\lg^2 x}$$

14)

$$f(x) = (x^2 + \lg x) \cdot \arctan x$$

$$f'(x) = \left(2x + \frac{1}{x}\right) (\arctan x) + (x^2 + \lg x) \left(\frac{1}{1+x^2}\right)$$

$$2x \arctan x + \frac{\arctan x}{x} + \frac{x^2 + \lg x}{1+x^2}$$

E 4

$$\lim_{x \rightarrow 0} \frac{x^3 \sin x}{x^2 + 2 \cos x - 2} = \frac{x^3 (x + o(x))}{x^2 + 2 \left(1 - \frac{x^2}{2} + o(x)\right) - 2} = \frac{x^4 + o(x^4)}{x^2 + 2 - x^2 + 2o(x^2) - 2} = \frac{x^4 + o(x^4)}{o(x^2)}$$

Traluppiamo ad un ordine più alto

$$f(x) = \sin x \quad P_2(x) = \sin 0 + x \cos 0 + \frac{(-\sin 0)}{2!} x^2 = 0 + x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$